MOI HIGH-KABARAK 2025 PREDICTION

Kenya Certificate of Secondary Education

KCSE PREDICTION SERIES 1

233/1 **Chemistry (Theory)**

Paper 1

Time: 2 Hours

NAME……………………………………………………………………………………………

SCHOOL……………………………………………… SIGN…………………………

INDEX NO……………………………………………… ADM NO………………………

Kenya Certificate of Secondary Education.

INSTRUCTIONS TO CANDIDATES

- *Write your name and admission number in the spaces provided.*
- *Answer **all** questions in the spaces provided*
- *KNEC mathematical tables and silent electronic calculators **may** be used for calculations.*
- *All workings **must** be clearly shown where necessary.*

FOR EXAMINERS USE ONLY

QUESTIONS	MAXIMUM SCORE	STUDENTS SCORE
1-29	80	

*Answer **all** questions in the spaces provided*

1. (a) Explain why aluminium is a better conductor of electricity than magnesium. **(1 mark)**

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(b) Other than cost and ability to conduct, give two other reasons why aluminum is used for making overheadelectriccables while magnesium is not. **(1mark)**

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2. In the Haber process, the industrial manufacture of ammonia is given by the following equation:

$$N_{2\,(g)} + 3H_{2\,(g)} \longrightarrow 2\,NH_{3(g)}$$

(a) Name one source of hydrogen gas used in this process. **(1 mark)**

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(b) Name the catalyst used in the above reaction. **(1mark)**

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(c) State any two uses of ammonia. **(1 mark)**

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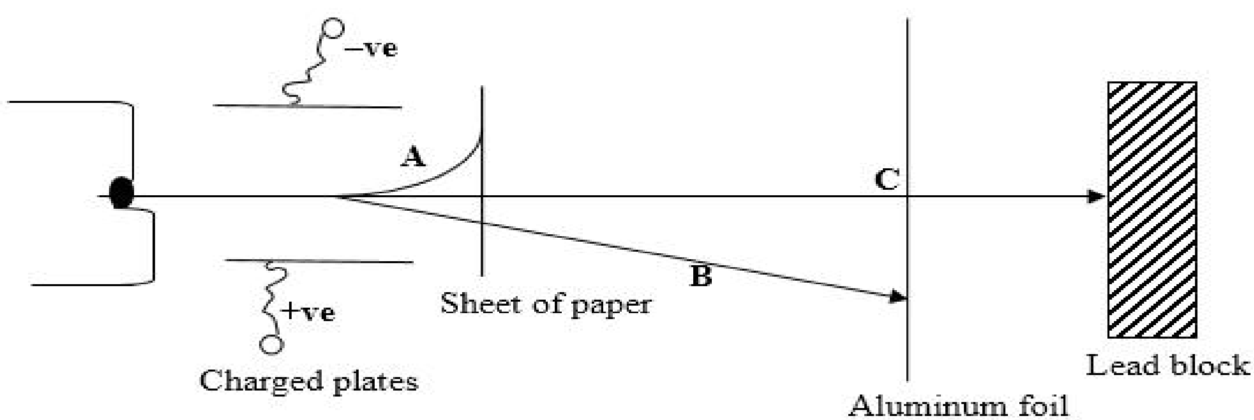
3. Use the diagram below to answer the questions that follow:



a) Identify the radiations. **(1 mark)**

A.………………………………………………………………………………………

C.………………………………………………………………………………………

b) The table below gives the rate of decay for a sample of radioactive element **K**. Study it and answer the question that follows:

Mass (kg)	Time(Days)
72	0
9	120

Determine the half-life of element K. **(2 marks)**

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4. The pH values of some solutions labeled **K,H,L,P and R** are given in the table **below**. Use the information to answer the questions that follow.

pH	8.0	14.0	1.0	6.5	7.0
Solution	K	H	L	P	R

(a)Identify the solution with the highest concentration of hydroxyl ions. **(1 mark)**

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(b)Which solution is likely to be sodium chloride solution? **(1 mark)**

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(c) Which solution would react most vigorously with magnesium metal? **(1 mark)**

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5.a) Name the class to which the following cleansing agents belong: **(1mark)**

i) $R-COONa$

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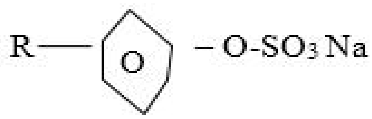
(ii) 

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b) Which cleaning agent between (i) or (ii) above is preferred for cleaning garments while using water from a dam containing dissolved calcium chloride? Explain **(2 marks)**

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6. Describe how crystals of sodium chloride can be prepared starting with $50cm^3$ of 2M sodium hydroxide solution. **(3marks)**

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7.a) Propane (C_3H_8) and Carbon(IV)oxide (CO_2) diffuses at the same rate under the same conditions. Explain. **(1 mark)**

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b) Propane is a hydrocarbon. What does the term hydrocarbon mean**?** **(1 mark)**

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8. The reaction between hydrochloric acid and potassium dichromate (VI) can be used to demonstrate a reversible reaction. The ionic equation is given below

$$2CrO_4^{2-}{}_{(aq)} + 2H^+{}_{(aq)} \rightleftharpoons Cr_2O_7^{2-}{}_{(aq)} + H_2O_{(l)}$$

Yellow Orange

Explain the observation that would be made when few drops of dilute hydrochloride acid is added to the equilibrium mixture. **(2marks)**

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9. a) Draw a diagram to show how an iron ring can be electroplated with pure silver. **(2marks)**

b) Give two reasons why electroplating is necessary.

(1 mark)

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10.(i) State **two** observations that can be made when burning magnesium is lowered in a gas jar full of chlorine. **(2 marks)**

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(ii) Write a chemical equation for the reaction that occurs in (i) above. **(1 mark)**

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11. The products formed by action of heat on nitrates of element X, Y and Z are shown below.

Nitrate of	Products formed
X	X oxide +Nitrogen (IV) Oxide + Oxygen
Y	Y +Nitrogen (IV) Oxide+ Oxygen
Z	Z nitrite + oxygen

a) Arrange the metals in order of increasing reactivity. **(1 mark)**

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b) Which element forms a soluble carbonate?

(1mark)

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c) Give an element that can beY. **(1 mark)**

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12.**3.0g** of an organic compound containing Carbon, hydrogen and oxygen only produced 4.4g of Carbon (IV) oxide and 1.8g of water on complete combustion.

a) Calculate its empirical formula. **(2 marks)**

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b) Calculate in molecular formula if its formula mass is 60. **(1 mark)**

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13. **100cm^3** of0.05M Sulphuric (VI) acid were placed in the flask and small quantity of potassium Carbonate added. The mixture was boiled to expel all carbon (IV) oxide. 25cm^3 of the resulting solution required 18cm^3 of 0.1M potassium hydroxide solution to neutralize it. Calculate the mass of potassium carbonate added. (K = 39, O = 16, C = 12) **(3 marks)**

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14. A sample of air contains nitrogen, oxygen and argon. Describe how oxygen gas can be obtained. **(3 marks)**

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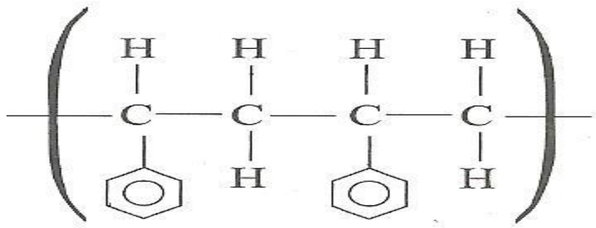
15. a) Draw the structures of the following compounds. **(2 Marks)**

i) Ethylbutanoate

ii) 3-ethyl – 3 – methyl hexane

iii) The structure below represents a portion of polyphenylethene.



Draw the structure of the monomer and name it. **(1mark)**

16. A compound whose general formula is $P(OH)_{3\,(s)}$ reacts as shown by the equation:

$$P(OH)_{3(s)} + OH^-_{(aq)} \longrightarrow P(OH)_4^-{}_{(aq)}$$

$$P(OH)_{3(S)} + 3H^+_{(aq)} \longrightarrow P^{3+}_{(aq)} + 3H_2O_{(l)}$$

(a) Name any two **elements** whose hydroxides behave like P. **(2 marks)**

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(b) What name is given to compounds which behaves like $P(OH)_3$ in the above two reactions. **(1 mark)**

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17. You are provided with the following electrode potentials of four half-cell reactions. Letters do not represent the actual symbol of the element.

$A^{2+}(aq) + 2e^- \longrightarrow A(s) \qquad E^\ominus = -0.14V$

$B^{3+}_{(aq)} + 2e^- \longrightarrow B(s) \qquad E^\ominus = +0.77V$

$C^{3+}_{(aq)} + 3e^- \longrightarrow C(s) \qquad E^\ominus = -0.20V$

$D_{2(g)} + 2e^- \longrightarrow 2D^-(aq) \qquad E^\ominus = +1.07V$

(a) **Identify** the strongest reducing agent. Explain. **(2marks)**

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(b) Calculate the emf of the two half cells that when combined would produce the largest emf. **(1 mark)**

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18. Use the following information to answer the questions that follow.

$CO_{(g)} + \frac{1}{2}O_{2(g)} \longrightarrow CO_{2(g)} \qquad \Delta H = -283kJmol^-$

$H_{2(g)} + \frac{1}{2}O_{2(g)} \longrightarrow H_2O_{(l)} \qquad \Delta H = -286kJmol^-$

$CH_3OH_{(l)} + 3/2O_{2(g)} \longrightarrow CO_{2(g)} + 2H_2O_{(l)} \qquad \Delta H = -715kJmol^-$

a) Draw an energy cycle diagram and label the various heat changes. **(2 marks)**

b) Use the energy circle above to calculate heat of formation of methanol. **(1 mark)**

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19. The solubility of iron (II) sulphate at 22°C is 15.65g/100g of water. Calculate the mass of iron (II) sulphate crystals in 90g of saturated iron (II) sulphate solution**.** **(2marks)**

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20. Which Use the information in the table below to answer the questions that follow.

(The letters do not represent the actual symbols of the elements).

Element	**Q**	**P**	**R**	**S**	**T**
Atomic number	18	5	3	5	20
Mass number	40	10	7	11	40

(a) Which **two** letters represent the same element**.**

1mk

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(b) Give the number of neutrons in an atom of element **R** **1mk**

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21. A solid sample was suspected to contain zinc (II) ions. Describe a systematic test to confirm the presence of zinc (II) ions. **(3 Marks)**

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22. During the electrolysis, a current of 2 amperes was passed through the copper (II) sulphatesolution for 4 hours. Calculate the volume of the gas produced at the anode. (1 F= 96500 C, MGV = 24000cm^3). **(3 marks)**

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23. A volume of 10cm^3 of ethene gas (C_2H_4) was exploded with 50cm^3 of oxygen.

(i) Write the equation of the reaction for the combustion of ethene. **(1mk)**

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(ii) Calculate the volume of gaseous mixture. **(2marks)**

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24.a) What is a fuel? **(1 mark)**

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b) Ethanol has a molar heatof combustion of -1360kJmol^{-1}. Calculate the heating value of ethanol.
(C=12,H=1,O=16) **(2 marks)**

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25.a) Why is the percentage of carbon (IV) oxide in the atmosphere fairly constant? **(1 mark)**

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(b) Calculate the volume of carbon(IV)oxide in 8,000 m^3 of air contained in a hall.**(2 marks)**

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26. **State two** conditions that would make the boiling point of water to be higher than 100^{o}C.

(2marks)

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27a) Name two substances that can be used in chemical test for water. **(1mark)**

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b) Describe how a student can determine the purity of tap water in a school laboratory **(2marks)**

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28. Sodium hydroxide can be prepared through the following methods: I and II.

I Sodium metal $\xrightarrow{\text{cold water}}$ Sodium hydroxide + Hydrogen gas

II Concentrated sodium chloride $\xrightarrow{\text{process A}}$ Sodium hydroxide + Chlorine gas + hydrogen gas

a) Name one precaution that needs to be taken in method I. **(1 mark)**

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b) Give the name of process A. **(1 mark)**

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c) Describe the chemical test of hydrogen gas.

(1 mark)

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29. i) What does CFCs mean? **(1 mark)**

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ii) State one environment effect of CFCs.

(1 mark)

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